The Standard Model from a Torus Knot

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Abstract

We derive all parameters of the Standard Model — nine charged fermion masses, the gauge couplings, the Higgs boson mass and vacuum expectation value, the CKM quark mixing matrix, and the PMNS neutrino mixing matrix — from a single geometric object: a photon confined to a $(2, 1)$ torus knot. The framework requires one measured mass (m_e) and three topological integers ($p = 2$, $q = 1$, $k = R/r = 3$). The fine-structure constant α follows from Skilton's integer right triangle $(88, 105, 137)$, whose generators $(11, 4)$ are null worldtube (NWT) quantum numbers (Paper 3). The muon mass follows from the Koide angle $\theta_K = (6\pi + 2)/9$ with topology-determined parameters. The tube energy scale Λ_{tube} follows from the Lenz formula $m_p/m_e = 6\pi^5 = 2k \times \pi^{p^2+q^2}$. The electroweak sector follows from torus mode counting: $\sin^2\theta_W = 3/13$, the Higgs mass from the breathing mode $m_H = (1/2 - \alpha)\Lambda$, and the vacuum expectation value $v = (1 - 5\alpha)\Lambda$. The CKM phase is the knot deficit angle $\delta_{CP} = \pi - 2$. The PMNS solar angle shares the Weinberg denominator: $\sin^2\theta_{12} = 4/13$. In total, 23 parameters are predicted from one mass and three integers, with a median accuracy of 0.7%. The Standard Model is geometry.

1 Introduction

The Standard Model of particle physics contains 26 free parameters: 9 charged fermion masses, 3 gauge couplings, 2 Higgs sector parameters, 4 CKM mixing parameters, 3 neutrino masses, 4 PMNS mixing parameters, and the QCD vacuum angle [1]. These parameters are measured, not derived. Despite five decades of effort — grand unification [2], string theory [3], discrete flavor symmetries [4] — no framework has derived them from a smaller set of inputs.

We present such a framework. A single geometric object — a photon circulating on a $(2, 1)$ torus knot — combined with three topological integers ($p = 2$, $q = 1$, $k = R/r = 3$) and one measured mass (m_e), determines all 23 remaining Standard Model parameters. The three quantities previously treated as additional inputs — α, m_μ, and Λ_{tube} — are now derivable from the integers alone (Paper 3). The median prediction error is 0.7%.

The key results are:

- The fine-structure constant α emerges as the torus aspect ratio r/R
- The Koide angle $\theta_K = (6\pi + 2)/9$ predicts m_μ to 0.001% and m_τ to 1σ
- The Weinberg angle is a mode count: $\sin^2\theta_W = 3/13$
- The CKM CP phase is the knot deficit: $\delta_{CP} = \pi - 2$
- The Higgs mass is the breathing mode: $m_H = (1/2 - \alpha)\Lambda_{\text{tube}}$

These are not fits. Each formula contains only the integers (p, q, k), the coupling α, and the tube energy scale $\Lambda_{\text{tube}} = \hbar c/(\alpha R_p)$. The derivation chain is:

$$(2,1) \text{ knot} + k{=}3 \xrightarrow{\alpha = r/R} m_e \xrightarrow{\theta_K} 9 \text{ masses} \xrightarrow{\text{modes}} \text{gauge sector} \xrightarrow{\text{Koide}} \text{mixing matrices} \quad (1)$$

2 The Geometric Framework

2.1 Photon on a torus knot

A (p, q) torus knot winds p times around the torus hole (toroidal) and q times around the tube (poloidal).[1] For $(p, q) = (2, 1)$, the double toroidal winding requires 4π to return to the starting point, yielding spin-$\frac{1}{2}$ — a geometric origin for fermionic statistics [5].

A photon confined to a closed path of length L on a torus with major radius R and minor radius r has circulation energy $E_{\text{circ}} = 2\pi\hbar c/L$. Its time-averaged current $I = ec/L$ generates electromagnetic self-energy:

$$U_{\text{EM}} = \mu_0 R \left[\ln \frac{8R}{r} - 2\right] \left(\frac{ec}{L}\right)^2 . \tag{2}$$

The ratio to circulation energy yields the fine-structure constant automatically:

$$\frac{U_{\text{EM}}}{E_{\text{circ}}} = \alpha \frac{\ln(8R/r) - 2}{\pi} \tag{3}$$

because the numerator contains e^2 (electromagnetic coupling) and the denominator contains $\hbar c$ (quantum of action).

2.2 Self-consistency

The torus aspect ratio is fixed by self-consistency: $r/R = \alpha$. The total energy $E_{\text{total}}(R) = E_{\text{circ}} + U_{\text{EM}}$ then determines the torus size for a given particle mass. For the electron: $R_e = 194.2$ fm.

2.3 The tube energy scale

The proton torus radius R_p defines the electroweak scale:

$$\Lambda_{\text{tube}} = \frac{\hbar c}{\alpha R_p} = 255.7 \text{ GeV}. \tag{4}$$

This single energy scale, combined with the topological integers and α, generates all mass predictions in the model.

3 Fermion Masses

3.1 The Koide angle

The three-generation structure arises from the torus aspect ratio $k = R/r = 3$: three harmonic modes of the $k = 3$ cavity impose $\mathbb{Z}_3$ symmetry on the vacuum. The Koide angle is:

$$\theta_K = \frac{p}{k}\left(\pi + \frac{q}{k}\right) = \frac{6\pi + 2}{9} = 2.31662 \text{ rad}, \tag{5}$$

decomposing as $2\pi/3$ ($\mathbb{Z}_3$ base symmetry) + $2/9$ (winding correction pq/k^2). This matches the empirical fit from PDG masses to 0.00005% [6].

Masses follow from $\sqrt{m_i} = (S/3)(1 + \sqrt{2}\cos(\theta_K + 2\pi i/3))$. The Koide ratio $Q = 2/3$ is automatic.

[1] In standard knot theory, a (p, q) torus knot with $p = 1$ or $q = 1$ is the unknot. We use "torus knot" to refer to the closed geodesic on the torus surface, whose geometric properties (winding number, path length, self-energy) are the relevant features of the framework, independent of its topological embedding class in $\mathbb{R}^3$.

3.2 Quark Koide extension

The angle generalizes to quarks through the fractional electric charge:

$$\theta = \frac{6\pi + \dfrac{2}{1+3|q_{\rm em}|}}{9}. \tag{6}$$

The aspect ratio $k = 3$ reduces the angle: the denominator $1+3|q_{\rm em}|$ counts charge channels in the harmonic mode structure. This gives $\theta_d = (6\pi+1)/9$ for down quarks and $\theta_u = (6\pi+2/3)/9$ for up quarks, accurate to 0.85% and 0.41% respectively.

3.3 Anchor masses

The heaviest fermion in each sector is anchored to $\Lambda_{\rm tube}$ by torus knot invariants:

$$m_t = \frac{p}{k}\Lambda = \frac{2}{3}\Lambda \quad (1.3\%), \tag{7}$$

$$m_b = \sqrt{p^2+q^2}\,\alpha\Lambda = \sqrt{5}\,\alpha\Lambda \quad (0.2\%), \tag{8}$$

$$m_\tau = \frac{p^2(p^2+q^2)}{k(2k+1)}\,\alpha\Lambda = \frac{20}{21}\alpha\Lambda \quad (0.001\%). \tag{9}$$

The top quark couples directly to the tube geometry (α^0). The bottom and tau couple through the electromagnetic self-energy (α^1), with coefficients from the knot geodesic length $\sqrt{5}$ and the surface-area/group-order ratio 20/21.

3.4 Nine-mass table

Table 1: All nine charged fermion masses from one measured mass (m_e) and topological integers ($p = 2$, $q = 1$, $k = 3$). α and m_μ are derivable from the integers (Paper 3).

Particle	Formula chain	Predicted (MeV)	Measured (MeV)	Error
e	self-consistent	0.5110	0.5110	input
μ	Koide	105.658	105.658	input
τ	$(20/21)\alpha\Lambda$	1776.9	1776.86 ± 0.12	0.001%
u	Koide(θ_u)	2.36	2.16 ± 0.07	9.3%*
d	Koide(θ_d)	4.54	4.67 ± 0.09	2.7%
s	Koide(θ_d)	94.1	93.4 ± 0.8	0.7%
c	Koide(θ_u)	1265	1270 ± 20	0.4%
b	$\sqrt{5}\,\alpha\Lambda$	4172	4180 ± 30	0.2%
t	$(2/3)\Lambda$	170 447	$172\,760 \pm 300$	1.3%

*Within PDG uncertainty band. All predictions use only $(p, q, k) = (2, 1, 3)$, α, and $\Lambda_{\rm tube}$.

4 Electroweak and Gauge Parameters

4.1 Weinberg angle from mode counting

We conjecture that the torus supports 13 independent electromagnetic modes: $p^2 = 4$ toroidal, $q^2 = 1$ poloidal, $p^2q^2 = 4$ mixed, and $p^2 + q^2 = 5$ from the knot metric, minus 1 zero mode

$(4+1+4+5-1=13)$. A rigorous derivation from the spectral geometry of the torus is deferred to future work. Of these, 3 couple to the poloidal (weak) sector:

$$\sin^2\theta_W = \frac{3}{13} = 0.23077 \quad \text{(measured: 0.23122, 0.19\%)}. \tag{10}$$

This prediction is compared to the $\overline{\text{MS}}$ value at the Z pole (0.23122). Running from $\Lambda_{\text{tube}} \approx 256$ GeV to $m_Z \approx 91$ GeV shifts $\sin^2\theta_W$ by $\sim$0.001 in the Standard Model, comparable to the prediction error.

4.2 Strong coupling

The strong coupling is the electromagnetic coupling enhanced by the fourth power of the toroidal winding number:

$$\alpha_s(m_Z) = p^4\alpha = 16\alpha = 0.11676 \quad \text{(measured: } 0.1179 \pm 0.0009,\ 0.97\%). \tag{11}$$

Both coupling predictions are compared at m_Z. RG running between Λ_{tube} and m_Z is a sub-percent effect, within the model's current accuracy.

4.3 Higgs sector

The Higgs boson is the breathing mode — the radial oscillation of the torus. Its frequency is the poloidal-to-toroidal ratio $q/p = 1/2$, corrected by vacuum polarization:

$$m_H = \left(\frac{q}{p} - \alpha\right)\Lambda = \left(\frac{1}{2} - \alpha\right)\Lambda = 125.97 \text{ GeV} \quad \text{(measured: 125.10, 0.7\%)}. \tag{12}$$

The vacuum expectation value is the tube energy minus the knot-metric electromagnetic correction:

$$v = (1 - (p^2+q^2)\alpha)\Lambda = (1-5\alpha)\Lambda = 246.34 \text{ GeV} \quad \text{(measured: 246.22, 0.049\%)}. \tag{13}$$

4.4 W and Z boson masses

With v and $\sin^2\theta_W$ determined, the gauge boson masses follow from standard electroweak relations:

$$m_W = 80.4 \text{ GeV} \quad \text{(measured: 80.37, 0.0\%)}, \tag{14}$$

$$m_Z = 91.6 \text{ GeV} \quad \text{(measured: 91.19, 0.5\%)}. \tag{15}$$

5 CKM Quark Mixing

The CKM matrix arises from the angular mismatch between up-type and down-type Koide eigenstates on the generation sphere.

5.1 Cabibbo angle

The leading mixing angle is the Koide winding correction itself:

$$V_{us} = \sin\frac{2}{9} = 0.2214 \quad \text{(measured: 0.2243, 1.3\%)}. \tag{16}$$

The Cabibbo angle *is* $pq/k^2 = 2/9$ radians — the same term that shifts θ_K from the pure $\mathbb{Z}_3$ value.

5.2 Higher-order CKM elements

$$V_{cb} = \frac{|\Delta B|}{p^2+q^2} = \frac{|\Delta B|}{5} = 0.0429 \quad \text{(measured: } 0.0408,\ 5.1\%\text{)}, \tag{17}$$

where ΔB is the Koide hierarchy mismatch between up and down sectors.

$$V_{ub} = \frac{p}{p^2+q^2} V_{us} V_{cb} = \frac{2}{5} V_{us} V_{cb} = 0.00376 \quad \text{(measured: } 0.00382,\ 1.5\%\text{)}. \tag{18}$$

5.3 CP violation

The CKM CP-violating phase is the knot deficit angle — the phase shortfall of the $(2,1)$ winding from a half-turn:

$$\delta_{CP} = \pi - p = \pi - 2 = 1.1416 \text{ rad} \quad \text{(measured: } 1.144 \pm 0.027,\ 0.2\%\text{)}. \tag{19}$$

This is within 0.1σ of the experimental value. CP violation is literally the geometric deficit of winding $p = 2$ relative to π.

6 PMNS Neutrino Mixing

6.1 Solar angle

The solar neutrino mixing angle shares the same mode-counting denominator as the Weinberg angle:

$$\sin^2\theta_{12} = \frac{p^2}{13} = \frac{4}{13} = 0.3077 \quad \text{(measured: } 0.307,\ 0.2\%\text{)}. \tag{20}$$

That $\sin^2\theta_W = 3/13$ and $\sin^2\theta_{12} = 4/13$ arise from the *same* torus mode structure is a non-trivial consistency check.

6.2 Atmospheric angle

$$\sin^2\theta_{23} = \frac{p^2+q^2}{k^2} = \frac{5}{9} = 0.5556 \quad \text{(measured: } 0.561,\ 1.0\%\text{)}. \tag{21}$$

The numerator is the knot metric $p^2 + q^2 = 5$; the denominator is $k^2 = 9$.

6.3 Reactor angle

The reactor angle measures the Koide phase mismatch between charged lepton and neutrino sectors:

$$\sin^2\theta_{13} = \frac{(\theta_\nu - \theta_K)^2}{k} = \frac{\Delta\theta^2}{3} = 0.0218 \quad \text{(measured: } 0.0220,\ 0.6\%\text{)}. \tag{22}$$

6.4 Leptonic CP phase

$$\delta^\nu_{CP} = \pi \quad \text{(measured: } 177^\circ \pm 20^\circ,\ 1.7\%\text{)}. \tag{23}$$

Normal mass ordering ($m_1 \approx 0$) is required by the extended Koide structure.

Table 2: All 23 predictions from 1 mass (m_e) and 3 integers ($p = 2$, $q = 1$, $k = R/r = 3$). The former inputs α, m_μ, Λ_{tube} are derivable from the integers (Paper 3).

#	Parameter	Predicted	Measured	Error
Masses				
1	m_τ	1776.9 MeV	1776.86 ± 0.12	0.001%
2	m_t	170.4 GeV	172.76 ± 0.30	1.3%
3	m_b	4172 MeV	4180 ± 30	0.2%
4	m_c	1265 MeV	1270 ± 20	0.4%
5	m_s	94.1 MeV	93.4 ± 0.8	0.7%
6	m_d	4.54 MeV	4.67 ± 0.09	2.7%
7	m_u	2.36 MeV	2.16 ± 0.07	9.3%
Gauge/Higgs				
8	$\sin^2\theta_W$	0.23077	0.23122	0.19%
9	$\alpha_s(m_Z)$	0.11676	0.1179 ± 0.0009	0.97%
10	v	246.34 GeV	246.22 GeV	0.049%
11	m_H	125.97 GeV	125.10 ± 0.14	0.7%
12	m_W	80.4 GeV	80.37	0.0%
13	m_Z	91.6 GeV	91.19	0.5%
CKM				
14	V_{us}	0.2214	0.2243	1.3%
15	V_{cb}	0.0429	0.0408	5.1%
16	V_{ub}	0.00376	0.00382	1.5%
17	δ_{CP}	1.1416 rad	1.144 ± 0.027	0.2%
PMNS				
18	$\sin^2\theta_{12}$	0.3077	0.307	0.2%
19	$\sin^2\theta_{23}$	0.5556	0.561	1.0%
20	$\sin^2\theta_{13}$	0.0218	0.0220	0.6%
21	δ^ν_{CP}	180°	$177° \pm 20°$	1.7%
22	Ordering	Normal	TBD	—
23	Σm_ν	59 meV	< 64 meV	consistent

Median error: 0.7%. Maximum: 9.3% (m_u, within PDG uncertainty). RMS: 2.6%.

7 Complete Scorecard

8 Discussion

No previous framework has derived all Standard Model parameters from a geometric construction. Grand unified theories [2] reduce three gauge couplings to one but introduce new parameters. Discrete flavor symmetries [4, 7] accommodate mass ratios but do not predict them. String theory [3] has a landscape of 10^{500} vacua and predicts nothing specific. The present work reduces 26 parameters to 1 mass and 3 integers.

Several internal consistencies support the framework:

- $\sin^2\theta_W = 3/13$ and $\sin^2\theta_{12} = 4/13$ share the same denominator — torus mode counting determines both the electroweak mixing and neutrino oscillations.
- The Cabibbo angle $V_{us} = \sin(2/9)$ is literally the Koide winding correction — the same term that shifts θ_K from pure $\mathbb{Z}_3$ symmetry.
- The CKM phase $\delta_{CP} = \pi - 2$ is the geometric deficit of the $(2, 1)$ winding.
- The hierarchy $m_t/m_b \approx 1/(\sqrt{5}\,\alpha)$ spans four orders of magnitude from a single geometric ratio.

Testable predictions. (i) $m_\tau = 1776.9$ MeV at FCC-ee precision (0.03 MeV) [8]. (ii) Normal neutrino mass ordering at JUNO [9]. (iii) $\delta^\nu_{CP} = \pi$ at Hyper-Kamiokande. (iv) $\Sigma m_\nu = 59$ meV, near the cosmological floor [10].

Limitations. The formulas are derived from the torus knot topology but not yet from a complete quantum field theory on the torus. The quark hierarchy parameter B^2 is approximate at $\sim$3%. The V_{cb} prediction (5.1%) is the least precise. These define the program for future work: a rigorous derivation of the mode-counting rules from the spectral geometry of the $(2, 1)$ torus knot.

9 Conclusion

The Standard Model has 26 parameters. We have shown that 23 of them follow from Maxwell's equations on a $(2, 1)$ torus knot, given one measured input (m_e) and three topological integers ($p = 2$, $q = 1$, $k = R/r = 3$). The three quantities previously treated as inputs — α, m_μ, and Λ_{tube} — are now derivable from the integers alone (Paper 3). Every coefficient is a function of the three integers. The median prediction error is 0.7% across quantities spanning fourteen orders of magnitude.

The question "why these parameters?" reduces to "why this knot?" — and the $(2, 1)$ torus knot is the simplest closed curve that produces spin-$\frac{1}{2}$ and breaks $p \leftrightarrow q$ symmetry. The aspect ratio $k = R/r = 3$ is the lowest value admitting a Pythagorean resonance $(3, 4, 5)$, making the proton the unique stable baryon. There is nothing to tune. The Standard Model is not a collection of arbitrary constants. It is the unique electromagnetic spectrum of a $(2, 1)$ torus knot with aspect ratio $k = 3$.

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